

Research Statement

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Koetter and Kschischang introduced the operator-channel framework for communication through subspaces over finite fields, modeling rank loss as erasures and interference as errors [1]. Soleymani and Mahdaviifar extended this framework to complex vector spaces through the analog operator channel, motivated by non-coherent wireless networks [2]. Information is encoded in a complex vector subspace: rank loss can reduce the transmitted subspace, while interference can add unwanted directions. The receiver must recover the transmitted subspace from this altered observation. My primary research is in algebraic methods for coding theory, especially the construction and analysis of subspace codes for the analog operator channel.

My current work with Felice Manganiello concerns two algebraic approaches to coding in this setting. One project studies character-polynomial codes, including their exact cardinality and parametrization. The other constructs complex subspace codes from finite-field subspace codes using extended multiplicative characters. Together, these projects examine how algebraic constructions and character maps affect code size, distance, and the relationship between messages and codewords.

Character-Polynomial Codes

Character-polynomial (CP) codes were introduced by Soleymani and Mahdaviifar as an algebraic construction for analog subspace coding [2]. They are one-dimensional complex subspace codes formed by evaluating finite-field polynomials and mapping their values to complex roots of unity using an additive character.

In our paper, *Exact Cardinality and Nonredundant Parametrization of Character-Polynomial Codes* [3], we identify redundancy in the usual polynomial description over extension fields: distinct polynomials can generate the same codeword. This redundancy arises because an additive character of an extension field factors through the field trace to its prime field, and therefore may not distinguish all finite-field evaluation values.

We analyze the trace-evaluation map using p -cyclotomic cosets and determine the exact number of distinct codewords. In the paper's standard evaluation setting, the cardinality is

$$|\mathcal{C}(\mathcal{F})| = p^{\sum_{\ell \in L} |C_p(\ell)| - 1},$$

where L is a set of representatives for the relevant p -cyclotomic cosets and $C_p(\ell)$ is the coset represented by ℓ . We also construct a refined set of defining polynomials whose map to the code is a bijection: every codeword has exactly one representative in this set. These results give an exact code-size count and a nonredundant polynomial parametrization, clarifying how polynomial messages correspond to codewords in rate calculations and future coding and decoding work.

Complex Subspace Codes from Finite Fields

In a second project, we construct complex subspace codes from finite-field subspace codes using an extended bijective multiplicative character [4]. The analog operator channel motivates this

construction: it allows us to bring established algebraic families over finite fields into the complex setting while controlling their code parameters.

First, we establish an exact relation between the subspace metric and the squared Frobenius distance between orthogonal projection matrices, the distance used in the analog-channel framework of Soleymani and MahdaviFar [2]. Within their error-and-erasure correction framework, we show that the subspace metric provides a guarantee at least as strong as the one based on squared projection distance, and can provide a strictly stronger guarantee. This comparison motivates our study of complex subspaces equipped with the subspace metric as a setting for error-correcting codes for the analog operator channel.

To construct codes, we fix a basis matrix for each finite-field codeword, apply the extended character entrywise, and take the complex row space of the resulting matrix. Since the character map is not additive, different basis choices can produce different complex realizations. For every choice, however, the construction preserves cardinality and codeword dimensions and does not decrease the minimum subspace distance.

We prove a stronger distance result for character image codes of lifted maximum-rank-distance (MRD) codes. Let $\mathcal{C}_R \subseteq \mathbb{F}_q^{l \times m}$ be an MRD code. Its lift consists of the subspaces $\text{rs}_{\mathbb{F}_q}(I_l \ M)$, where $M \in \mathcal{C}_R$. Using these systematic basis matrices, the character image code of the lift is the lift of the character image code of $\mathcal{C}_R$:

$$\chi^*(\text{Lift}(\mathcal{C}_R)) = \text{Lift}(\chi(\mathcal{C}_R)).$$

We prove that the character image code of a lifted MRD code has the same cardinality and minimum subspace distance as the original finite-field lifted MRD code.

The cardinality equality follows from the general character construction. For the distance result, we study $\chi(\mathcal{C}_R)$ as a rank-metric code with entries in μ_{q-1}^0 , the alphabet consisting of zero and the relevant roots of unity, with rank computed over $\mathbb{C}$. A Singleton-type bound for these rank-metric codes, combined with the MRD cardinality and the rank comparison between $\mathcal{C}_R$ and $\chi(\mathcal{C}_R)$, forces equality of their minimum rank distances. The lifting identity $d_S = 2d_R$ in [6] then gives equality of minimum subspace distances.

We also derive a Singleton-type bound for complex subspace codes in the fixed systematic lifting chart. Specifically, consider codes whose codewords have the form $\text{rs}_{\mathbb{C}}(I_l \ M)$, where $M \in (\mu_{q-1}^0)^{l \times m}$. For such a code $\mathcal{C}$, the bound is

$$|\mathcal{C}| \leq |\mu_{q-1}^0|^{m \left(l - \frac{d_S(\mathcal{C})}{2} + 1 \right)}.$$

We prove that character image codes of lifted MRD codes attain this bound, so they have maximum cardinality within this specified family for their minimum subspace distance. This result applies to lifted MRD codes generally. The Koetter–Kschischang code is one example, since it is obtained by lifting a Gabidulin code [1, 5].

Future Directions

A primary next step is to develop decoding methods for character image codes of lifted MRD codes, with the character image code of the Koetter–Kschischang code as one concrete case. The equality of minimum subspace distances establishes a precise error-correction capability, but does

not itself provide a decoder. Standard Gabidulin and lifted-MRD decoders use finite-field algebraic structure, which the character map does not preserve: in general, χ is not additive. Moreover, a received complex subspace need not itself be the character image of a finite-field subspace. These decoders, therefore, do not transfer directly to the complex image codes.

One direction is to develop nearest-codeword algorithms under the complex subspace metric that use the lifting form and the finite alphabet. Another is to develop algebraic decoding methods that use the underlying MRD structure while accounting for the nonadditivity of the character map. I also want to understand how broadly this construction applies to other well-known finite-field subspace codes, and how basis choices affect the resulting complex realizations and their decoding.

My broader interests include commutative algebra, algebraic number theory, and computational algebraic geometry, particularly where these subjects connect with algebraic coding theory. These are areas I hope to explore alongside my current work. My long-term goal is to contribute to algebraic coding theory through constructions with precise parameter guarantees and algebraic structure that can support decoding over complex spaces.

References

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